ZOO-514 SUBJECTIVES

FINAL TERM- 100%

PRINCIPLES OF ANIMAL LIFE-I

GUESS PAPER 2024

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CURRENT SUBJECTIVES

Reproductive Patterns in Animals:

All invertebrates and vertebrates have one of the three reproductive patterns.

1= Oviparous: These are the animals which lay eggs and development takes place outside the body of animals. Fertilization may be internal or external. Majority of invertebrates and many vertebrates are oviparous.

2= Ovoviparous: These are the animals which retain eggs into the body but the young one gets all its food from yolk of egg. Both fertilization and development is internal. Many members of annelids, brachiopods, gastropods, fishes and reptiles.

3= Viviparous: These are the animals in which eggs develop into the body and the young one gets its nourishment from mother. Both fertilization and development is internal. Scorpions, most mammals, elasmobranch fishes, few amphibians and reptiles are viviparous.

Types of variables:

Parameters in experiment that may change is called Variables.

1. **Independent Variable:** This is the parameter manipulated or tested in the experiment. In the example, the independent variable is the daylight length, as it is being tested to observe its effect on the reproductive readiness of mice.
2. **Dependent Variable:** This is the parameter measured to determine the outcome of the experiment. In the example, the frequency of mating in mice is the dependent variable, and its variation is observed based on changes in the independent variable (daylight length).
3. **Control Variable:** This is a parameter that is kept constant to prevent its influence on the results. In the example, temperature is a control variable, as it is maintained constant to ensure that any observed changes in mating frequency are attributed to variations in the independent variable (daylight length).

How regulation occur in aquatic animals

Aquatic animals regulate water differently based on their environment. Freshwater animals, in hypotonic environments, conserve ions to prevent loss. Marine animals, in hypertonic environments, manage water to avoid salt buildup and dehydration. Each adapts to maintain internal balance amid varying external conditions.

What is the Biological magnification?

Biological magnification is the process where the concentration of certain substances, like toxins or pollutants, increases as they move up the food chain. Organisms at higher trophic levels accumulate more of these substances, posing risks to top predators and humans consuming contaminated species.

How decomposer help in nutrient recycling process.

Decomposers, mainly bacteria and fungi, play a crucial role in nutrient recycling. They break down dead organic matter into simpler compounds using extracellular enzymes. By doing so, decomposers release locked minerals, prevent environmental clutter, and recycle essential nutrients, ensuring the sustainability and balance of the ecosystem.

Applications of zoology in medical research:

Zoology plays a crucial role in medical research by contributing to the understanding of diseases caused by animals. Through techniques and protocols, scientists diagnose diseases and elucidate the underlying mechanisms. For instance, in the case of dengue, understanding how Aedes mosquitoes transmit the disease is vital. Zoological insights aid in designing preventive measures and medications. Moreover, zoology facilitates medical advancements by studying the effects and mode of action of new drugs on animal models before their application in human medicine, ensuring safety and efficacy.

Difference between ectosymbiosis and Endosymbiosis:

Ectosymbiosis:

It is a symbiosis in which symbionts live on the body surface of the host or in the digestive tract. E.g., Lice, barnacles and cleaner fish

Endosymbiosis:

It is the relationship in which one symbiont live within the tissue of other. E.g., nitrogen fixing bacteria in root nodules, algae inside the corals.

Describe the threatens to biodiversity:

Biodiversity of all natural areas is threatened mainly due to habitat loss, acid rain, pollution, urbanization and agriculture. Human is exploiting 40% of earth net primary production.

Function of placenta:

Placenta is s temporary organ which connects developing fetus to mother via umbilical cord. Human have placenta formed form chorion and allantois. The tiny chorionic villi sent out from blastocyst form placenta. Fetus is fed through placenta. And primarily functioning as provide Nutrient and oxygen exchange, waste elimination, and hormone production during pregnancy.

Hormones during birth:

During birth, hormones play vital roles. Relaxin, produced by the ovaries and placenta, helps separate the mother's pelvic bones for easier passage of the baby. Adrenocorticotropic hormone (ACTH) from the baby's pituitary gland stimulates adrenal glands to release prostaglandins, working with oxytocin. Oxytocin, triggered by uterine wall stretch, causes powerful contractions. These hormones collectively facilitate the birthing process, ensuring the expulsion of the baby and later the placenta in a sequence called afterbirth.

What are the steps of fertilization:

Fertilization involves the fusion of egg and sperm, forming a diploid zygote. There are two types: external (in water) and internal (within an animal's body). It can occur through copulation, cloacal connection, or via spermatophore. The vitelline membrane surrounds the egg, with receptors for sperm binding. Sperm carry lysins in their acrosomal head, dissolving the egg membrane. Only one sperm successfully fertilizes the egg, and egg activation follows, involving biochemical changes to complete fertilization and initiate embryonic development.

What are the types of Epithelial tissues:

Epithelial tissues, covering and lining various body parts, have diverse types based on shape and layers. Squamous epithelium is single-layered, flat, found in lungs, heart, and kidneys. Cuboidal epithelium, cube-shaped, is in kidney tubules and glands, facilitating secretion and absorption. Columnar epithelium lines the digestive tract, gallbladder, aiding absorption and enzyme secretion. Pseudostratified ciliated columnar epithelium, with cilia, propels mucus in bronchi and uterus. Stratified squamous epithelium, with many layers, protects skin, esophagus, mouth, and vagina from abrasion.

Define entropy with examples:

Things tend become more stable by acquiring low energy state (disordered). The measure of degree of disorganization is called entropy. For example, when natural gas burns to give heat energy, some of it cooks food, other released into environment. Unusable energy is increasing entropy.

Define keystone species:

Those species on which several other species are dependent are called keystone species. Removal of keystone species will cause the death of dependents.

Describe Female reproductive structure of female body.

Reproductive role of human female is more complex. The main functions of human female reproductive system are egg production, to nourish, carry and protect developing embryo. Female also nurse the young one after birth.

Human female reproductive system consists of ovaries, uterine tubes fimbriae and uterus, vagina, external genitalia and mammary glands. Ovaries produce eggs. They are also responsible for estrogen and progesterone production. Uterine tubes carry eggs from ovaries to uterus. Fimbriae are finger like projections which receive the egg into uterine tube. Uterus is the place where embryo is implanted for development. Vagina receives sperms during sexual activity. External genitalia are important in sexual arousal and have protective role while mammary glands are to feed the milk to newborn.

Difference between arctic tundra and alpine tundra ecosystems.

Arctic tundra is in the northern hemisphere with temperatures from -30°C to 20°C, hosting the Nganasan and Nenets people. Animals include arctic fox, polar bear, and caribou. Alpine tundra, at high elevations globally, is too cold and windy for extensive plant life. Both have lichens, grasses, and shrubs, but alpine tundra flora has more roots and rhizomes.

Importance of sexual reproduction:

Sexual reproduction is essential for the genetic diversity and adaptability of species. It results in offspring with unique combinations of genes from two parents, fostering variation within populations. This diversity enables better adaptation to changing environments, enhances resistance to diseases, and promotes overall species resilience. Sexual reproduction also facilitates the elimination of harmful genetic mutations through natural selection. The complexity of sexual reproduction contributes to the evolutionary success and ecological balance of diverse organisms across ecosystems.

Enlist ten organs in human body:

1= Heart, 2= Lungs, 3= Liver, 4= Kidneys, 5= Brain, 6= Stomach, 7= Intestines (small and large), 8= Pancreas, 9= Spleen, 10= Skin.

Enlist location of the conifers forest in Pakistan:

Coniferous forest are dominated by cone bearing, needle shape evergreen plants called conifers. They have cool summer and cold winter. In Pakistan they are present at 1000 to 4000 altitude. Coniferous forests are present in Chitral, Swat, Dir, Abbot Abad, Hilly Areas of Balochistan, Murree and Azad Kashmir. Coniferous trees are used in buildings and furniture. So, these forest are seriously damaged by human. Climatic changes are affecting the coniferous forests by mild, short winter and acid rain.

Ecological Hierarchy:

It is the way in which ecological components interact with each other • There are five levels • Organism • Population • Community • Ecosystem • Biosphere.

Components of male reproductive system:

Important components of male reproductive system include scrotum, testes, rete testis, epididymis, duct deferens and ampulla, Seminal Vesicle.

Binary fission & multiple fission:

1= Binary fission is a process where one organism divides into two, producing approximately equal-sized offspring. It can be asymmetrical, transverse, or longitudinal. Asymmetric fission occurs in organisms like amoeba, dividing from any side; transverse fission involves cleavage across an organism, as seen in paramecium, and longitudinal fission divides an organism lengthwise, as in planaria and euglena.

2= Multiple fission involves successive divisions of the nucleus followed by cytoplasmic division, resulting in one cell producing many cells. Amoeba and tapeworm exhibit multiple fission, with amoeba forming cysts under unfavorable conditions and generating new amoebae through this process.

Plants life in grasslands:

Grasslands support a diverse range of plant life adapted to the open and often seasonally dry environment. Plant life is dominated by different types of grasses making layers. Plant life include clover, salvia, wheat, barley, coneflowers and many species of grasses such as, purple needle grass, wild oats, foxtail, ryegrass, and buffalo grass. Grasslands support large populations of herbivores and carnivores including insects, deer, zebra, wildebeest, giraffe, bison, lions, tigers and hyenas. Important grasslands include Great Plains of North America, Pampas of South America, Veldt of South Africa, Steppes of Central Eurasia, and surrounding the desert of Australia. Continuous global warming has great impact on grasslands. Grasslands are under extensive pressure due to conversion to agricultural land and urban development.

Primary and secondary embryonic induction:

The capacity of some cells to evoke developmental process in others is called embryonic development. It is wide spread phenomenon in development. Spemann and mangold first reported embryonic induction. When the dorsal lip of blastopore was transplanted to lateral or ventral side of other embryo, it developed notochord, somite, ectoderm of host and formed the whole system formed. Dorsal lip of blastopore can induce the development of whole embryo, it is called as primary organizer. The phenomenon of induction by blastopore is called primary induction. Formation of other type of cell by inductive process is termed as secondary induction.

Polysaccharide and it characteristics:

A polysaccharide is a complex carbohydrate composed of multiple sugar units linked together. Examples include starch, glycogen, and cellulose. Polysaccharides serve as energy storage (starch and glycogen) and structural components (cellulose) in organisms. They consist of long,

branched or linear chains of monosaccharide units, providing diverse functions crucial for cellular processes, energy release, and structural integrity in living organisms.

Define Vertical stratification:

Vertical stratification refers to the layering of different vegetation levels in a forest or ecosystem, from the forest floor to the canopy. This stratification creates distinct ecological niches, with specific plant and animal species adapted to each layer's unique conditions.

Mosaic development and regulative development:

Mosaic Development:

- When a particular blastomere is isolated form embryo, it forms characteristic structure.
- In the absence of that cell the structure is missing.
- Mosaic development is characteristic of most protostomes.

Regulative Development:

- In many animals, fate of cell depends upon interaction with neighbors rather than cytoplasm itself.
- Each cell is able to produce entire embryo.
- If a blastomere is removed, the other change their fate to compensate this.

Define Parasitoids:

These are the animals that deposit eggs or spend developmental stages on the other animals. • As development proceed they kill the host. • They have properties of predator and parasite • E.g., parasitoid wasps.

Blastula formation in echinoderms:

Cleavage results in a ball of cells called morula. Morula is a solid ball of cells without any cavity. At about 120 cell stage, a fluid filled cavity blastocoel is formed in the embryo. There is a single layer of cells around the blastocoel. Embryo at this stage is called blastula. At blastula stage embryo comes out of the fertilization membrane.

Mimicry and it's types:

Mimicry is the phenomenon in which one species gains resemblance with other species and gets protection from predators. This resemblance may be with one or more than one species. 1= In batesian mimicry, one species resembles with other species and protected by aposematic coloration. Flies belonging to family Syrphidae resemble with bees or wasps, thus protected. Here flies are mimic and bee or wasps are model. 2= Mulerian mimicry both mimic and model both are distasteful. If predator eats any one of these, it will not prey again any one of them.

Importance of carbohydrates as energy source and support

Carbohydrates are essential biomolecules comprising sugars, starches, and fibers, serving as a primary energy source for living organisms. Structurally, they consist of carbon, hydrogen, and oxygen atoms. Simple carbohydrates, like glucose, provide quick energy, while complex carbohydrates, found in grains and vegetables, offer sustained energy. Additionally, carbohydrates play a crucial role in cellular communication and contribute to the structural integrity of cells and tissues, making them indispensable for overall biological functions.

Plans of division in amphibian:

Amphibians have mesolecithal eggs. Cleavage completely divide cell in two cells. Due to yolk at vegetal pole cleavage is slow. Size of cells at animal pole is smaller while larger at vegetal pole. First cleavage start from animal to vegetal pole. Cleavage at vegetal pole is slow due to large amount of yolk. It also divides gray crescent from middle. Second cleavage is perpendicular to the first and divides embryo into four. Third cleavage is equatorial and right angle to first two. Separate animal pole and vegetal pole. Fourth cleavage is meridional. Cleavage lines pass in

such a way that cells lie exactly one above the other (radial cleavage). This radial cleavage is the characteristic of deuterostomes.

Morulation: Cleavage results a ball of cells called morula. Morula is a solid ball of cells without any cavity. Cell towards animal pole are smaller while larger at vegetal pole due to presence of yolk.

Blastulation: After morula formation, a fluid filled cavity blastocoel is formed in the embryo. Cell form multiple layers around the blastocoel. In amphibians blastocoel is formed towards animal pole due to yolk at vegetal pole.

How knowledge of zoology helps in improving environment.

Knowledge of zoology is crucial for preserving biodiversity and ecosystems. It enables understanding of animal roles in maintaining environmental balance, identifying bioindicators of pollution, and developing conservation strategies. Zoological insights aid in safeguarding habitats, preventing extinction, and appreciating the aesthetic and ecological value of diverse animal species for a healthier environment.

Difference between determinate cleavage and indeterminate cleavage:

Determinate cleavage results in cells with predetermined fates early in development, restricting their potential. In contrast, indeterminate cleavage produces cells with undetermined fates, remaining totipotent and capable of forming a complete organism. Indeterminate cleavage is common in early embryonic stages, allowing greater flexibility in cell differentiation.

Difference between primordial and primary follicles.

During first five months of development many small follicles are formed in the ovary of female fetus. These are called primordial follicles. At the time of puberty during each menstrual cycle some primordial follicles began to develop further and become primary follicles which then converted to secondary follicles and graafian follicle.

Explain the steps of mesoderm in amphibians gastrulation:

In amphibian gastrulation, mesoderm formation involves cells moving from the surface to the presumptive mesoderm region. Gastrulation begins with the blastopore opening between the gray crescent and ventral pole. Cells roll over the dorsal lip, undergo involution, and migrate, eventually closing the blastopore, forming the mesodermal layer in amphibian embryos.

define Assimilation of nitrogen:

Assimilation of nitrogen refers to the incorporation of nitrogen into organic compounds by living organisms. Plants absorb nitrogen primarily in the form of nitrate or ammonium from the soil. Once assimilated, nitrogen compounds become part of essential biomolecules, such as amino acids and proteins. Animals obtain assimilated nitrogen by consuming plants or other organisms. Microorganisms in the soil also play a crucial role in converting organic nitrogen into forms that plants can assimilate.

Explain feature of fertilization and development in amphibians:

In amphibians, fertilization typically occurs externally in water. The sperm fertilizes the egg, forming a zygote. The embryos undergo holoblastic cleavage, forming blastula and gastrula stages. Gastrulation involves the movement of cells to create germ layers. Amphibians often have indeterminate cleavage, allowing cells to have varied fates. The process includes the formation of structures like the notochord. Development may involve a larval stage, such as tadpoles, which later undergo metamorphosis into adult forms, adapting to terrestrial life.

Development stages in amphibain

Amphibian development stages include fertilization, forming a zygote, followed by cleavage, blastula, and gastrula stages. Embryos undergo gastrulation, forming germ layers. Some species

have larval stages, like tadpoles, which metamorphose into adults, adapting from aquatic to terrestrial environments.

Oligotrophic and eutrophic:

Oligotrophic and eutrophic describe the nutrient levels in aquatic ecosystems. Oligotrophic environments have low nutrient concentrations, often resulting in limited plant growth. Eutrophic environments, on the other hand, have high nutrient levels, promoting abundant plant growth and potentially leading to issues like algal blooms and oxygen depletion.

Define Gross productivity net productivity:

When an animal eats, food passes through its gut and nutrients are absorbed. Total amount of energy assimilated by an organism when an animal eats food is called gross productivity. The amount of energy stored in the form of biomass is called net productivity.

Survivorship curve and types with example:

Survivorship refers to the probability of an individual's survival to a certain age. There are three types of survivorship curves: Type I, where most individuals survive to old age (humans); Type II, with a constant mortality rate throughout life (birds); and Type III, where high mortality occurs early in life, and survivors tend to live longer (many fish species). These curves help describe the distribution of mortality in a population over different life stages.

Herpetology and malacology:

Herpetology: Herpetology is the branch of zoology that deals with the study of amphibians and reptiles.

Malacology: Malacology is the branch of invertebrate zoology that deals with the study of the mollusca.

What is the structure of sperm:

Sperm, the male reproductive cell, consists of a head, containing the nucleus with genetic material, and an acrosome filled with enzymes to penetrate the egg. The midpiece has mitochondria for energy production, and the tail, or flagellum, propels the sperm for fertilization.

Which system is formed by endodermis layer of germ layer?

The endoderm layer of the germ layer gives rise to the endodermal tissues and organs, forming the endodermis system. This system includes structures such as the lining of the digestive and respiratory tracts, liver, pancreas, and other internal organs derived from the endodermal germ layer during embryonic development.

Differentiate between intertidal and neritic ecosystem:

The Intertidal Zone extends from splash zone ocean to low tide marks. This region is subjected to the action of both tides and waves. It is the harshest and the richest of all marine environments. Here is pounding surf, sun, wind, rain, erosion, sedimentation and temperature fluctuation. This zone has bounty of habitat and plenty of nutrients. Here, plants are larger.

The Neritic Ecosystem extends from littoral zone to the edge of continental shelf. It is also called the shallow water. Nutrients are delivered by rivers and upwelling at the edge of continental shelf. This region is more productive than open sea. Planktonic and attached algae contribute to the primary productivity.

Threats to environmental resources and what should be done to overcome this:

Environmental resources face threats like deforestation, pollution, and climate change. To address this, sustainable practices must be adopted, promoting reforestation, waste reduction, and renewable energy. Global cooperation on environmental policies, stricter regulations, and public awareness are crucial. Conservation efforts, habitat protection, and responsible resource use must

be prioritized to ensure a sustainable future, balancing human needs with the preservation of biodiversity and ecosystem health.

Classification of egg on the base amount of yolk:

Eggs are classified based on the amount of yolk into three types. Microlecithal eggs have minimal yolk, common in echinoderms. Mesolecithal eggs, with moderate yolk, are found in amphibians. Telolecithal eggs, abundant in yolk, are typical in birds and reptiles. Yolk quantity influences embryonic development; microlecithal eggs undergo holoblastic cleavage, while telolecithal eggs experience meroblastic cleavage, forming a blastoderm atop the yolk. These classifications correlate with the reproductive strategies and developmental needs of different species.

Male reproductive system:

Male reproductive system have two basic function: Production of sperms and Transfer of sperm to female reproductive tract. Important components of male reproductive system include scrotum, testes, rete testis, epididymis, duct deferens and ampulla. Seminal Vesicle. Male reproductive system consists of a pair of testes enclosed in a sac like structure called scrotum. Each testes has a large number of seminiferous tubules. These are the sites for sperm production. Seminiferous tubule join a network of tubules called rete testis. After that there is a dilated tubule called epididymis. Epididymis has three functions: 1) storage and maturation of sperms. 2) Smooth muscles that help to propel sperm during ejaculation and 3) Transfer sperms to duct deferens. Duct deferens transfer comes out of scrotum and passes through the lower part of abdominal wall via inguinal canal. Sometime this point is weakened and intestine part may protrude out. This condition is called inguinal hernia. Around urinary bladder duct deferens enlarges to form ampulla. Next to ampulla is called ejaculatory duct. After combining with urinary tube it moves in penis as urethra. There are different glands present in reproductive system. Important glands are seminal vesicles, prostate gland and bulbourethral gland.

Types of data with examples long.

The information obtained from observations is called data. Data is of two types:

Quantitative data involves numerical values, such as height, weight, and temperature, allowing for precise measurements and statistical analysis. Qualitative data lacks numerical values and includes characteristics like odor, shape, and color, providing descriptive information. Both types contribute to a comprehensive understanding of phenomena, with quantitative data offering precision, and qualitative data providing richer context and insights into the non-numeric aspects of observations.

Composition of cytoskeleton:

The cytoskeleton is a cellular framework composed of protein filaments. Three main components include microtubules, made of tubulin, providing structural support and aiding in cellular transport; microfilaments, consisting of actin, involved in cell shape and movement; and intermediate filaments, contributing to cell structure and mechanical strength.

Define metabolism and its role in living organism:

Metabolism refers to the sum of all chemical processes within a living organism that sustain life. It involves two main components: catabolism, breaking down complex molecules into simpler ones to release energy, and anabolism, building complex molecules from simpler ones, consuming energy. Metabolism fuels essential functions such as growth, repair, and energy production, maintaining cellular structures and supporting life processes. It is a dynamic and regulated system that enables organisms to adapt to environmental changes and maintain internal balance.

Holoblastic Cleavage:

The pattern of cleavage in which egg is completely divided into two cells. Egg contain little or moderate amount of yolk. Cleavage in fertilized eggs of Amphibians, mammals, echinoderms, flatworms and nematodes.

Meroblastic Cleavage:

The pattern of cleavage in which egg is not completely divided into two cells. This is due to large amount of yolk. In these embryo development takes place around or on the top of yolk. e.g., cleavage in fertilized eggs of birds and reptiles.

Camouflage:

It is the phenomenon in which the colour pattern of animal helps to hide the animal from others. There are two major types of camouflage: cryptic coloration and countershading.

Sexual reproduction:

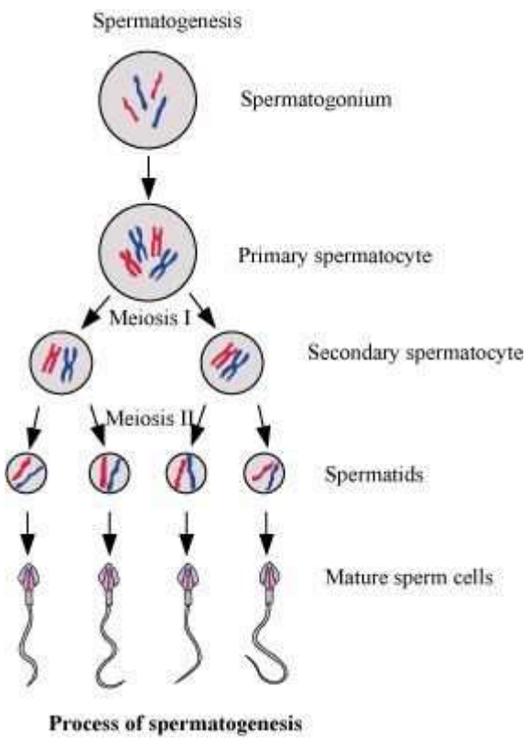
Sexual reproduction is a biological process where two parent organisms contribute genetic material to produce offspring with genetic variation. This process involves the fusion of gametes, typically sperm and egg, leading to the formation of a new individual.

Impact of increasing population on resources?

Increasing population strains resources, leading to overconsumption, environmental degradation, and resource depletion. Pressure on water, food, energy, and land exacerbates ecological and social challenges, impacting biodiversity, climate change, and societal well-being. Sustainable practices are crucial to mitigate these adverse effects.

Spermatogenesis explain with flow chart:

Spermatogenesis is the process of sperm cell development in the testes. It begins at puberty and involves mitotic divisions (spermatogonia to primary spermatocytes), meiosis I (primary spermatocytes to secondary spermatocytes), and meiosis II (secondary spermatocytes to spermatids). Spermatids undergo structural changes, forming mature spermatozoa. The process results in four functional sperm cells per spermatogonium. This flow chart illustrates the stages of spermatogenesis, detailing the transformation of germ cells into mature sperm for reproductive function.



Process of spermatogenesis

Formation of mesoderm and endoderm in chick?

In the chick embryo, mesoderm and endoderm formation occur during gastrulation. The process begins with the migration of cells from the primitive streak to form the three primary germ

layers. Mesoderm arises from the middle layer, while endoderm originates from the inner layer. This spatial differentiation establishes the foundation for the development of various tissues and organs. The complex movements and interactions during gastrulation are critical for proper embryonic patterning and subsequent organogenesis in the chick embryo.

Function of progesterone:

Progesterone, a hormone produced by the ovaries, supports pregnancy by preparing the uterine lining for embryo implantation. It maintains the endometrial environment and inhibits further ovulation, essential for maintaining a conducive environment for embryonic development during the early stages of pregnancy.

Biome with examples:

A biome is a large geographical region characterized by distinctive climate, flora, and fauna. Examples include the tropical rainforest with dense vegetation, the arctic tundra with cold temperatures, and the desert biome with sparse plant life and extreme temperatures.

Difference btw torpor and hibernation:

Torpor and hibernation are states of reduced metabolism to conserve energy. Torpor is a short-term, daily period of inactivity and lowered body temperature, commonly observed in hummingbirds or bats. Hibernation, in contrast, is an extended period of dormancy during winter to survive extreme conditions, with a more profound decrease in metabolic rate. It decreased metabolism from weeks to months • The set point of thermoregulatory center drop to 20oC Hibernation is typically seen in animals like bears, hedgehogs, or ground squirrels adapting to seasonal changes.

Define the structure of Mitochondria:

Mitochondria are double-membraned organelles with an outer and inner membrane. The inner membrane forms folds called cristae, increasing surface area. The matrix, enclosed by the inner membrane, contains enzymes for cellular respiration. Mitochondria house their own genetic material (DNA) and replicate independently. These structures enable mitochondria to produce ATP through oxidative phosphorylation, making them vital for cellular energy production and contributing to various cellular functions in eukaryotic organisms.

Define mineralization and oxidation in sulfur cycle?

Mineralization in the sulfur cycle involves the conversion of organic sulfur compounds into inorganic forms like sulfate by microbial activity. Oxidation refers to the process of converting reduced sulfur compounds into their oxidized forms, releasing sulfur compounds into the environment, often mediated by bacteria.

How pH and temperature affect enzyme?

Enzyme activity is influenced by pH and temperature. Each enzyme has an optimal pH and temperature for activity. Deviations from these values can alter enzyme structure, affecting substrate binding and reaction rates. Extreme pH or temperature can denature enzymes, rendering them inactive.

Monozygotic Twins:

Monozygotic Twins are from single zygote and are identical twins. They are strikingly alike and of same sex. About 33% have separate placenta. Majority have separate amniotic sac. Small proportion have one placenta and one amniotic sac which indicate separation occur after 9 day of pregnancy. There is conjoining risk, the condition is Siamese twining.

Dizygotic Twins:

Dizygotic Twins are produced from two zygotes. They are called as fraternal twins. They do not have striking similarities and may be of same or different sex. They have separate placenta and separate amniotic sac.

PAST SUBJECTIVES

Protogynous & Protandrous:

Sequential hermaphrodites have one sex in first phase of life and opposite sex in other phase of life. Means if animal is male in first phase of life, it will be female in second half of life. Those sequential hermaphrodites which are female in first phase of life and male in second phase of life called protogynous (femle first) e.g., wrasses. Those animals which are male in first phase of life and female in second phase of life are called protandrous e.g., clone fish.

Decomposers:

These are the organisms which obtain energy by breaking down dead organic matter into simpler compounds. They are also called saprotrophs. Chief decomposers of an ecosystem are bacteria and fungi. They release extracellular enzymes which breakdown substrate molecules. Organisms at all trophic level when die are used by decomposers. If there are no decomposers mounts of dead bodies will be everywhere.

Isotonic Solution: When solute concentration is same outside and inside of the cells. Water molecules move inside and out of the cell at same rate.

Hypertonic Solution:

When solute concentration is higher outside the cells than inside. Water molecules move out and the cell shrinks. This condition is called crenation.

Hypotonic Solution:

When solute concentration is lower outside the cell than inside. Water molecules move into the cell which causes the cell to swell and burst.

Reproduction & its Importance

Reproduction is the process of producing new individuals. It is the basic feature of all forms of life. It is essential for existence of species and also for continuity of life on earth. Capability to reproduce is called reproductive potential. All the animals have great reproductive potential. If resource are unlimited a single species of fishes may fill the all oceans.

Animals reproduce by two basic means: Asexual reproduction and sexual reproduction

Four types of nucleotide

(ap log isy short kr k b yad kr skty hain but easy wording he aik bar daikhny sy hi yad ho jay ga agr na ho to bs name yd kr lyen in k adenine, thymine, cytosine, guanine bs)

Nucleotides are the building blocks of nucleic acids like DNA and RNA. Each nucleotide consists of three components: a phosphate group, a pentose sugar, and a nitrogenous base. There are four types of nucleotides, differing in their nitrogenous bases:

1. **Adenine (A):** A nucleotide containing adenine as its nitrogenous base. Adenine pairs with thymine (in DNA) or uracil (in RNA) through hydrogen bonds.
2. **Thymine (T):** Present in DNA, thymine pairs with adenine through hydrogen bonds.
3. **Cytosine (C):** A nucleotide containing cytosine as its nitrogenous base. Cytosine pairs with guanine through hydrogen bonds in both DNA and RNA.
4. **Guanine (G):** A nucleotide containing guanine as its nitrogenous base. Guanine pairs with cytosine through hydrogen bonds.

In RNA, thymine is replaced by uracil (U), so the four types of nucleotides in RNA are adenine, uracil, cytosine, and guanine. These nucleotides link together to form the backbone of DNA and RNA molecules, encoding genetic information.

Define fragmentation

Fragmentation is the process of asexual reproduction in which an animal breaks into two or more pieces and each piece is capable of growing into a new organism. Fragmentation is common in different groups of animals including cnidarians, Platyhelminthes and echinoderms

Difference b/t Sprouters and Seeders

There are two types of plants: **sprouters and seeders**.

Sprouters are dormant meristem, storage and intact root help to regenerate quickly while **seeders** are long lived seeds, germination induced by fire.

Plants include toyon, chamise, poison oak, scrub oak, yucca and sumac. Important **animals** are insects, rodents, rabbits, lizards, snakes and mule deer.

Deference Between Smooth and Rough Endoplasmic Reticulum?

The smooth endoplasmic reticulum (SER) lacks ribosomes on its surface, while the rough endoplasmic reticulum (RER) possesses ribosomes. The SER is involved in lipid synthesis, detoxification, and storage. In contrast, the RER is primarily responsible for protein synthesis and modification. Both organelles contribute to cellular functions and maintain distinct roles.

Where does coexistence occur?

Coexistence: It is the phenomenon in which competing species live together

How Coexistence Occurs? Coexistence can occur when species utilize resources in slightly different way. MacArthur studied 5 species of warblers that use same caterpillar from different region of trees

Budding

Asexual reproduction in which an outgrowth appears on the body of animal which grows into complete organism. e.g., hydra, obelia and sponges.

In budding, a group of cells rapidly divides and form a small outgrowth called bud. This bud increases rapidly and form a complete organism. The bud may get detach from the parent or sometimes it may remain attached with the parental organism and form a closely associated group of individuals called a colony. Members of a colony belong to same species.

In sponges internal buds are produces which are called as gemmules. Gemmule is a collection of cells surrounded by body wall. When parent dies, each gemmule gives rise to a new individual.

Lentic Ecosystem

This is the ecosystem of standing water. Lentic ecosystem include lakes and ponds. Vertical stratification is common especially in temperate region. In summer, upper layer is warm while lower layer is cold and more dense, so no mixing. In fall, upper layer is dense so mixing of water occurs. Photosynthesis mainly in upper layer of water as light is filtered by water. Phytoplankton are main producers of lentic ecosystem. Plant life include phytoplankton, periphyton algae, emergents, floating- leaves, submersed and free floating plants. Animal life include zooplankton, invertebrates like crabs, crayfish and shrimps, molluscs while vertebrates mainly includes variety of fishes, reptiles and birds.

Lotic Ecosystem

It is the aquatic ecosystem of flowing water. It includes brooks, streams and rivers. Attached aquatic plants and algae provide some of the primary productivity. Most of the primary production comes from the surrounding plants by falling their materials during autumn and rainfall. Lotic systems typically have high oxygen level. Turbulence of water while flowing over rocks, logs and other objects causes oxygenation. This support the animal life in lotic system. Animals include different aquatic invertebrates, larvae of insects, fishes, amphibians reptiles birds and mammals.

What is the kingdom:

Kingdom is the highest rank of classification. All the living organisms are distributed into five kingdoms i.e., monera, protista, fungi, plantae and animalia. All the animals are included in kingdom aminalia.

Microtubules

Microtubules are hollow, slender, cylindrical structures. They are largest cytoskeletal elements. Each microtubule is about 25nm in diameter and made up of tubulin protein. Microtubules help in movement of organelles and chromosomes.

They are part of transport system (e.g., secretory vesicle movement) in the cell and are involved in overall shape changes of cell during specialization. Microtubules make different structures like mitotic apparatus, cilia, flagella and centrioles.

Oceanic ecosystem

Oceanic Ecosystem starts from continental shelf. It is the open sea. As it is nutrient poor so, it is relatively unproductive. Virtually all primary productivity is in the upper water. Phytoplankton in the upper water are the main producers of oceanic system. Invertebrate animals include worms, echinoderms and molluscs. Vertebrates include fishes from small fishes to whales and sharks. At depth no light so the animals here have luminescent organs which serve as torch light. These animals depend on nutrient coming from upper layers.

Evolutionary science and experimental science:

1= Evolutionary science explores the ultimate causes behind biological phenomena, focusing on the historical and evolutionary forces shaping traits and behaviors over time. For example, understanding why birds developed migratory behavior involves evolutionary science.

2= Experimental science, on the other hand, addresses proximate causes by investigating immediate factors affecting biological processes. Examples include experiments on maintaining body temperature or studying the process of cell division in a specific timeframe.

Define Isomerism. Give suitable example of carbohydrates

Molecules with same molecular formula but different structural formula. For example, glucose and fructose are isomers, both having the molecular formula C6H12O6 but differing in structure and properties.

Write three functions of epididymis of male reproductive system

Epididymis: Each testes has a large number of seminiferous tubules. These are the sites for sperm production. Seminiferous tubule join a network of tubules called rete testis. After that there is a dilated tubule called epididymis.

Epididymis has three functions:

- Storage and maturation of sperms.
- Smooth muscles that help to propel sperm during ejaculation and
- Transfer sperms to duct deferens.

Describe Coral Reefs

Coral Reef is the underwater ecosystem. They are formed by colonies of stony corals.

- Coral are members of class Anthozoa.
- They grow in association of zooxanthellae.
- Polyps of corals are held together by calcium carbonate.
- Reefs are formed over thousands of years in warm, clear and relatively shallow waters.
- The greatest depth for their growth is 50-60 m.

- ❖ Below this light is insufficient to support zooxanthellae.
- ❖ Coral Reefs are the richest source of biodiversity in oceans.
- ❖ Coral reefs occupy less than 0.1% of oceans, yet they are home of more than 25% marine species.
- ❖ Residents include sponges, cnidarians, nematodes, crustaceans, molluscs, echinoderms, tunicates and fishes.

What do we study in zoology?

Animal life is very fascinating to human being. It includes a great variety of animals ranging from microscopic invertebrates to very large size vertebrates. Beautiful domestic pets, gorgeous wildlife and stunning aquatic animals are all studied under the heading of zoology. When we see living organism's different questions rise in our mind like; How and where they live? What they eat? What they do? The answer to all questions is given by observing the animals living in natural environment, studying their activities (feeding, growth and reproduction) structure and function of different body parts up to molecular level.

Describe Intertidal Zone

The Intertidal Zone extends from splash zone ocean to low tide marks. This region is subjected to the action of both tides and waves. It is the harshest and the richest of all marine environments. Here is pounding surf, sun, wind, rain, erosion, sedimentation and temperature fluctuation. This zone has bounty of habitat and plenty of nutrients. Here, plants are larger.

They include seagrass, red algae and mangroves. Motile animals like fishes and crabs move with tides. Other such as amphipods, isopods, ghost crabs and polychaetes burrow in the sand and mud. Water trapped in tidal pools support a rich animal life such as echinoderms, barnacles, limpets and chitons. Attached animals close shells to avoid drought during low tides.

Morula cleavage / Morulation

Morulation: Cleavage results a ball of cells called morula. Morula is a solid ball of cells without any cavity. Cell towards animal pole are smaller while larger at vegetal pole due to presence of yolk.

Difference between loose and fibrous connective tissue

There are two general types of tissues: Loose connective tissues and hard connective tissues.

1. **Loose connective tissues** are strong flexible fibers of collagen which are interwoven with elastic fibers. They are located in epithelia of skin and throughout body. They wrap and cushion organs.
2. In **fibrous connective tissues** collagen fibers densely packed, parallel. They are located in dermis of skin; submucosa of digestive tract; fibrous capsule of organ and joints. Their major function is to provide structural strength.

Difference between radial and spiral cleavage

Spiral Cleavage: It is pattern in which cells of animal pole are not exactly laying above the vegetal pole cells. Cleavage takes place at oblique angle forming a spiral pattern. **Example**: protostomes cleavage

Radial Cleavage: It is pattern in which cells of animal pole are exactly laying above the vegetal pole cells. Upper tier of cells is exactly on the lower tier of cells. **Example**: deurostomes cleavage

Importance of nucleotides:

1. **Building Blocks of Genetic Material:** 1= Nucleotides form the structural basis of nucleic acids (DNA and RNA), containing genetic information crucial for cellular

functions. 2= Comprising a nitrogenous base, pentose sugar, and phosphate group, they establish the genetic code essential for storage and transmission of genetic information.

2. **Energy Currency in Cells:**1= Nucleotides, particularly nucleoside triphosphates (e.g., ATP, GTP, CTP, UTP), serve as high-energy molecules. 2= These molecules provide energy for diverse cellular processes, including metabolism, protein synthesis, cell division, and intracellular movement.

3. **Cell Signaling:** 1= Specific nucleotides like cGMP and cAMP function as secondary messengers, participating in cell signaling pathways. 2= They regulate cellular processes, contributing to the coordination and control of various physiological activities within cells.

4. **Cofactors for Enzymatic Reactions:** 1= Nucleotides serve as cofactors for enzymes in metabolic pathways. 2= Important coenzymes such as coenzyme A, FAD, FMN, NAD, and NADP+ contain nucleotide components, facilitating enzymatic reactions crucial for cellular metabolism.

5. **Role in Immune Function:** 1= Nucleotides contribute to the development and activation of immune cells, including natural killer cells. 2= These cells play a vital role in defending the body against infections and diseases, highlighting the immunological significance of nucleotides.

6. **Impact on Memory and Cognitive Function:** 1= Dietary nucleotides have been associated with improvements in memory and cognitive abilities. 2= Studies suggest a potential role for nucleotides in influencing neural processes, demonstrating their broader impact on physiological functions beyond cellular and molecular realms.

Difference between synchrony and asynchrony

Cleavage refers to cell divisions that occur during embryonic development. Initially all the cells of embryo divide at the same time this is called synchrony. As development proceeds, all cells don't divide simultaneously. This is called asynchrony. Eggs with evenly distributed yolk have cleavage pattern that result uniform size blastomeres. Egg with unevenly distributed yolk result in unequal size blastomeres.

Mineralization and oxidation in sulfur cycle:

The process by which organic sulfur is converted into inorganic form i.e., hydrogen sulfide, elemental sulfur and sulfide minerals is called mineralization and the step at which sulfides and elemental sulfur are converted to sulfates is called oxidation. Sulfates are reduced by plants, fungi and bacteria and incorporated.

Difference between renewable and non-renewables resources.

Natural resource can be divided into two main types: Renewable resources and Non-renewable resources. **1= Renewable resources** are those resources which are constantly available. After using they are reasonably replaced or recovered e.g., water, sunlight, air and biomass. They have minimal environmental impact, contribute to cleaner energy, and play a crucial role in addressing climate change. **2= Non-Renewable Resources** are those which once exhausted are no more available. They are not replaced or recovered e.g., oil, gas and minerals. Their utilization contributes to pollution, habitat destruction, and climate change.

Importance of phosphorus:

- Phosphorus is an essential component of DNA, RNA, ATP, Phospholipids, Vertebrate skeleton and Teeth.
- Sedimentary rocks contains deposits of phosphorus.
- Extraction of phosphorus takes place by weathering, rain and other erosion processes.
- Plant, microbes and fungi absorb phosphorus from soil.

- Animals use phosphorus by utilizing plants.
- Some phosphorus also comes through water.
- As plants and animals die, phosphorus is return back to the environment via decomposition.
- Living organisms may again use this phosphorus.
- Some phosphorus may run along with water and become part of sediments.
- Human has significant impact by using phosphorus in fertilizers.
- Agricultural runoff takes phosphorus to water bodies which excessive growth of algae. This phenomenon is called eutrophication.

Common terms related to reproduction

- Fertilization: Fusion of egg and sperm, forming a zygote.
- Gestation: Pregnancy period in mammals from conception to birth.
- Puberty: Adolescence stage with sexual maturation.
- Ovulation: Release of mature egg from ovary in females.
- Embryo: Developing stage after fertilization, before birth/hatching.

Explain asexual reproduction

It is the type of reproduction in which offspring are produced without fertilization of gametes. In asexual reproduction the basic method of cell division is mitosis which involves no genetic recombination. So, the animals produced by asexual reproduction are identical to their parents and siblings. Asexual reproduction is common the common method of reproduction in protozoa (Amoeba, paramecium and euglena) and lower invertebrates (hydra, jellyfish, and flatworms).

Define Inductive and deductive reasoning:

1= Inductive Reasoning: Inductive reasoning involves drawing general conclusions from specific observations or evidence. It moves from the particular to the general and is probabilistic rather than absolute. For example: After observing numerous instances of a cat, one may inductively reason that all cats observed so far have fur. The conclusion is likely but not guaranteed to be universally true.

2= Deductive Reasoning: Deductive reasoning starts with general premises or principles and uses them to reach a specific, logical conclusion. It is characterized by certainty, as the conclusion necessarily follows from the premises. For example: All humans are mortal (premise 1). Socrates is a human (premise 2). Therefore, Socrates is mortal (conclusion). The conclusion is certain based on the given premises.

Density-independent Factors

These factors affect population regardless of their number

- Weather conditions
- Earth quake
- Pollutants
- Deforestation
- Hurricane
- Fire

What are the advantages of asexual reproduction:?

Asexual reproduction offers advantages such as rapid population growth, as there is no need for a mate. It ensures genetic consistency, allowing successful preservation of favorable traits. This method is energy-efficient and well-suited for stable environments.

Neritic ecosystem

The Neritic Ecosystem extends from littoral zone to the edge of continental shelf. It is also called the shallow water. Nutrients are delivered by rivers and upwelling at the edge of continental shelf. This region is more productive than open sea. Planktonic and attached algae contribute to the primary productivity. Plants include kelps, algae, seaweeds, seagrasses and phytoplankton. Invertebrate animals include crabs, sponges, echinoderms, corals, and polychaetes whereas, vertebrate fauna include different types of fishes such as code, tuna, herring, capelin and many other species.

Structure of DNA:

The structure of DNA is a double helix formed by two polynucleotide chains. Each chain consists of nucleotides, which are composed of a sugar molecule (deoxyribose), a phosphate group, and one of four nitrogen-containing nucleobases: adenine (A), cytosine (C), guanine (G), or thymine (T). The two chains are held together by hydrogen bonds between the nucleobases. Adenine forms a complementary base pair with thymine, and cytosine pairs with guanine. The shape is attributed to the work of James Watson and Francis Crick, who proposed the double helix model in 1953, revolutionizing our understanding of genetics and molecular biology.

Hierarchy of classification

The Hierarchy of classification of animals is as follow:

- Kingdom
- Phylum
- Class
- Order
- Family
- Genus
- Species

The smallest group of classification system is species

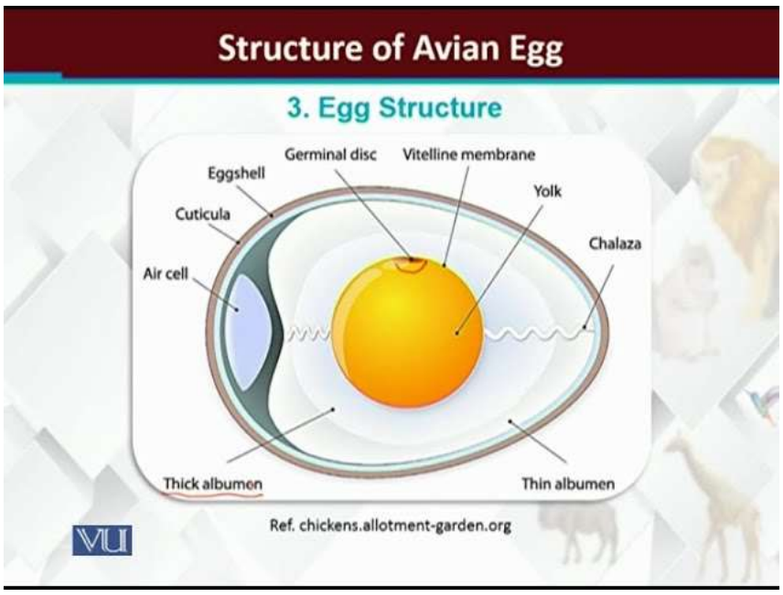
Structure of Avian Egg

Membranes and fluids are collected around the egg while moving through oviduct. Water and albumin is deposited around the egg. This protect from desiccation and mechanical damage.

Albumin is used for protection and a source of nutrition. Two denser strands of albumin called chalaza are attached to the egg shell. They suspend egg in the center of albumen. Shell gland secrete $CaCO_3$ which deposit around the egg.

It is also impregnated with proteins. Thousands of tiny pores (40-50μm) are for exchange of gases. Inside the shell two membranes are present. At the round end of egg an air pocket is present between two membranes.

Air pocket enlarges with time. Near the time of hatching the chick breaks air pocket, the air inflates its lungs and the chick start breathing.



Explain Fission:

Fission (Lenten word fissio, the act of splitting) is a type of asexual reproduction in which one organisms splits into two or more organisms.

There are two basic types of fission: Binary fission and multiple fission. The process in which one organism divides into two organisms is called **binary fission**. Both are approximately equal size. Binary fission may be asymmetrical, transverse and longitudinal.

Asymmetric organisms like amoeba may divide into two from any side, such fission is called asymmetric fission. When a line of cleave transversely divide an organism into two is called transverse fission. e.g., binary fission in paramecium.

When an organism is divided longitudinally into two daughter organisms, this is called longitudinal fission. e.g., binary fission in planaria and euglena.

Multiple fission is a number of successive divisions of nucleus followed by cytoplasmic division in which one cell divides into many cells. e.g., amoeba, tapeworm. Amoeba form cyst under unfavorable conditions and form new amoebae by multiple fission.

Define intermediate filaments

Intermediate filaments are heterogeneous group of cytoskeletal fibers. They are made up of more than 70 different types of proteins. They are of intermediate size (10nm diameter). Intermediate filaments help to maintain cell shape, maintain spatial arrangement of organelles and promote mechanical activities in cytoplasm.

Define Theory of Chemical evolution / Theory of cell evolution:

1= The Theory of Chemical Evolution proposes that life originated from non-living matter through gradual chemical processes. It suggests that simple organic molecules formed spontaneously, leading to the development of more complex compounds and eventually the emergence of life. This theory posits that prebiotic conditions on early Earth facilitated the synthesis of essential biomolecules, such as amino acids and nucleic acids.

2= On the other hand, the **Theory of Cell Evolution** focuses on the step-by-step progression from simple, self-replicating molecules to the first cellular entities. It explores how these primitive cells evolved, diversifying and acquiring the ability to harness and process energy, ultimately leading to the diverse array of life forms we observe today. Both theories contribute to our understanding of the origin and evolution of life on Earth.

Nitrogen cycle

- Nitrogen fixation converts atmospheric nitrogen into forms usable by plants.

- Plants take up nitrogen compounds from soil for growth.
- Decomposition returns nitrogen to soil; nitrification converts it to nitrate.
- Denitrification releases nitrogen back into the atmosphere.
- Human activities like farming impact the nitrogen cycle through fertilizers.

Characteristics of biome found in small patches / THE CHAPARRAL

It is the biome fount in small patches in most of the continents. It is characterized by very hot and dry summer with temperature up to 37.5oC. Winter is very mild and usually about 10oC and most of the rain occur in winter. Annual precipitation is 30-75cm.

Soil is thin, rocky, nutrient poor, highly susceptible to erosion and contains low amount of moisture. Plants of chaparral are low and shrubby. They have tough and waxy leaves which are broad leaves and evergreen. Plants have no thick bark so not very resistant however their recovery is very fast.

There are two types of plants: sprouters and seeders. Sprouters are dormant meristem, storage and intact root help to regenerate quickly while seeders are long lived seeds, germination induced by fire. Plants include toyon, chamise, poison oak, scrub oak, yucca and sumac. Important animals are insects, rodents, rabbits, lizards, snakes and mule deer.

Chaparral is represented by west coast of United States and South America, South Africa, western tip of Australia and coastal area of Mediterranianean.

Effect of pesticides on children vs. adults

- Children's developing bodies are more sensitive to pesticides.
- Children consume more relative to their weight, leading to higher pesticide exposure.
- Pesticides can impact brain development and behavior in children.
- Developing hormonal systems make children susceptible to pesticide effects.
- Health issues from early-life exposure can have lifelong impacts.

Factors of abiotic stress methods

Biotic Factors: All living organisms in ecosystem make biotic factors.

1= Producers, 2= Consumers, 3= Decomposers

Abiotic Factors: All nonliving things in ecosystem make abiotic factors.

1= Temperature, 2= Moisture, 3= Light, 4= Soil, 5= Air, 6= Water, 7= Fire.

Natural and artificial resources

Natural Resources:

- Natural resources are elements and materials found in the environment that are valuable to humans for various purposes.
- They are derived from the Earth's natural processes and include categories like minerals, water, forests, air, wildlife, and fossil fuels.
- Natural resources are essential for human survival and economic development, supporting industries such as agriculture, energy, and manufacturing.
- They can be classified into renewable resources (e.g., solar energy, wind energy) and non-renewable resources (e.g., oil, coal) based on their ability to replenish over time.

Artificial Resources:

- Artificial resources, also known as man-made or synthetic resources, are products created by humans through technological processes.
- They include items like manufactured goods, electronics, synthetic materials (plastics, synthetic fibers), and infrastructure (roads, buildings).

- ❖ Artificial resources are often derived from natural resources but transformed into new forms with added value through human intervention.
- ❖ The production and utilization of artificial resources can have significant environmental and sustainability implications.

Food chain & its Importance

It is the sequence of organisms in an ecosystem where energy moves from one organism to the next organism. It tells us the sequence to eat and being eaten.

Food chain is important:

1)To understand the feeding relationships between organisms.

2)To explore the energy flow mechanisms and matte circulation.

3)To understand the problems of biomagnification.

Food chains are mainly of two types: grazing food chain and detritus food chain.

Factors Affecting Reproductive Potential

- ❖ Number of offspring produced
- ❖ Survival to reproductive age
- ❖ Reproductive period duration
- ❖ Length of time taken to maturity
- ❖ Advantages of asexual reproduction

What is the effect of pollution on ecosystem

Biodiversity Impact: Pollution disrupts habitats and food chains, causing a decline in biodiversity. Sensitive species can be harmed or driven to extinction due to polluted environments.

Water Contamination: Industrial, agricultural, and urban pollutants contaminate water bodies. Toxic chemicals and excess nutrients harm aquatic life and disturb ecosystem balance.

Air Quality Degradation: Emissions from industries and vehicles release harmful pollutants into the air. Poor air quality affects both human health and the well-being of plants and animals.

Habitat Disruption: Pollution contributes to the degradation and destruction of natural habitats. Species' survival is compromised, leading to potential extinctions and ecosystem imbalance.

Climate Change Connection: Pollutants like greenhouse gases contribute to climate change. Altered weather patterns and temperatures disrupt ecosystems and species distribution.

Photoautotrophs and chemoautotrophs

Photoautotrophs use sunlight as energy source for the synthesis of food from inorganic compounds. They include green plants, algae and photosynthetic bacteria. Photosynthesis transform sunlight into chemical energy.

Out of total sunlight reaching to earth 30% reflects back, 20% is absorbed by atmosphere, 50% absorbed by earth and less than 1% of sunlight is used by photosynthesis.

Chemoautotrophs obtain energy by oxidizing inorganic compounds. They usually live in extreme environments. e.g., sulfur bacteria.

Why are estuaries important to our environment?

Estuaries are nurseries for crustaceans, molluscs and marine fishes. They are the feeding grounds for wading and swimming birds. Human have drained and filled estuaries. Estuaries are polluted by oil exploration and drilling. Acids rain have also threaten the productivity of estuaries.

Oogenesis:The process of sperms production is called **spermatogenesis** while the process of eggs production is called **oogenesis**.